



AEGIS Architecture

Anonymization & Exchange Gateway for Imaging Studies

SENDING SOURCES External Browser Upload · DICOM Network (C-STORE SCP, port 11112) · Programmatic Batch Ingest

React Upload Portal (PWA)

- Drag-and-drop: files, folders, DICOM directories
- Client-side tag de-identification (DICOM PS3.15)
- Anonymization preview — before / after tag diff
- Multi-study detection with per-study upload cards
- Per-project retained-tag anonymization profiles
- Auto-retry: 3x exponential backoff per file

DIMSE Receiver (pynetdicom)

- C-STORE SCP listening on port 11112
- Receives studies from PACS systems / scanners
- Institution attribution (AE title or IP CiDR)
- Calls POST /api/ingest on DICOM association close
- Durable retry queue + dead-letter (disk-persistent)
- Exponential backoff · operator control API

Key Privacy Principle

PHI is stripped in the browser BEFORE data leaves the hospital.

- Only tag-de-identified DICOM is transmitted
- HIPAA Safe Harbor — 18 identifier types removed
- Deterministic UID hashing + date shifting
- Zero software install required at sending site
- DIMSE path ingests raw files (internal-only path)

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Cloud Armor (DDoS / WAF) + Global HTTPS Load Balancer + Identity-Aware Proxy (admin routes) + Managed SSL cert v3 (api · admin · aegisimaging.ai · www) + Cloud Run (API · Dashboard · Landing · 8 sidecars)

API Backend (Go / Cloud Run)

- Upload orchestration (sessions, chunked PUT)
- Study / project / institution management
- Routing rules engine → auto-pipeline dispatch
- DICOMweb proxy (QIDO-RS + WADO-RS)
- Export shares: token-auth, ZIP download
- Email digest scheduler + SMTP relay
- DIMSE retry control proxy
- MCP tool backend + batch import CLI
- distroless image — minimal CVE surface

Admin Dashboard + OHIF (React / Cloud Run)

- Behind Identity-Aware Proxy (IAP)
- Study browser: filter, search, paginate, bulk ops
- OHIF Viewer — before/after defacing side-by-side
- 7-stage pipeline visualization per study
- RBAC: admin (write) + viewer (read-only)
- Routing rules, institutions, anon profiles
- Protocol templates, API keys, webhooks
- Audit log + CSV export, share management
- Federation peers, project lifecycle

Cloud SQL (PostgreSQL 15)

- Studies, projects, institutions, admin users
- Routing rules, audit trail, export shares
- Private IP · Secret Manager credentials
- Point-in-time recovery (7-day retention)

Cloud Storage (GCS)

- dicom/raw/{uid}/ — tag-de-identified
- dicom/clean/{uid}/ — defaced + processed
- bids/{uid}/ — NIFTI / BIDS output
- Shared volume (Go API + all 8 sidecars)

Processing Sidecars (Python / Cloud Run) — dispatched async by Go API, runs hands-free via routing rules

Defacing (Python)

- Head MRI / PET / CT — facial feature removal
- mri_reface / DeepDefacer / mri_deface backends
- nibabel fallback for dev / testing only
- Writes defaced DICOM to clean/ store
- SSIM-based QA score per study

PHI Detection (Python)

- Burned-in text OCR on DICOM pixel data
- Tesseract (local) / Cloud Vision / Textract
- Per-file findings with confidence scores
- Per-project confidence threshold config
- Informational — non-blocking to pipeline

QC Automation (Python)

- File integrity + required DICOM tag check
- Slice consistency (rows / cols / pixel spacing)
- SNR estimation (signal mean / corner noise)
- Coverage completeness by body part
- Missing slice gap detection

Classification (Python)

- Fills modality + body_part from DICOM headers
- Heuristic: tags → SOP UID → series description
- Cloud Vision / Rekognition image fallback
- Re-evaluates routing rules after classify
- Confidence threshold gating (≥ 0.5)

BIDS Conversion (Python)

- DICOM → NIFTI via dcm2nix
- BIDS-compliant directory structure
- sub-{hash8}/anat | func | dwi | perf | pet/
- JSON sidecar metadata per series
- Privacy: UID-hashed subject labels
- ZIP download via Go API

Protocol Check (Python)

- Verifies TR / TE / flip / thickness vs template
- Classic + Enhanced DICOM (multi-frame)
- Per-project protocol templates with CRUD
- numeric / exact / range / contains_all match
- critical / warning / info severity levels

Synth MRI (Python) ★

- Synthetic brain MRI generation
- CPU: Shepp-Logan phantom (numpy + pydicom)
- GPU: MONAI BraTS LDM (INCLUDE_MONAI=true)
- T1w contrast, Rician noise, face anatomy
- Seeded + reproducible, 20 – 200 slices
- Imports via normal pipeline (auto-dispatch)

Security Layers

- VPC private subnets + Cloud NAT
- Cloud Armor DDoS / WAF
- TLS 1.2+ on all endpoints
- CMEK (Cloud KMS)
- IAM least privilege
- Secret Manager (DB creds, API keys)
- Cloud Audit Logs
- Artifact Registry vuln scanning
- distroless containers (Go API)
- IAP on all admin routes
- No PHI in email / audit entries

Export & Sharing

Export Portal (React / public SPA)

- Token-authenticated share links
- Study info, expiry countdown (server-anchored)
- ZIP download of approved DICOM files

DICOM Forwarding (route_to)

- DICOMweb STOW-RS destinations
- DIMSE C-STORE to remote AE Title
- Auto-forwards on study approval

Share Lifecycle

- Extend / revoke anytime
- Immutable download log per share
- Export analytics dashboard

MCP Server + AI Agent Tools

Model Context Protocol (Claude)

- 17 read tools: studies, audit, stats, health
- 13 write tools (confirm:true + reason required)
- Zod-validated schemas — read-safe by default
- DIMSE retry proxy: process, replay, dead-letter

Batch Import + Webhooks + API Keys

- aegis-import CLI — bulk historical migration
- 5 webhook events, HMAC-SHA256 signed payloads
- Machine-to-machine bearer tokens (hashed)

Landing Page (nginx / Cloud Run)

- aegisimaging.ai — public marketing site
- React + Vite built, served by nginx
- Interactive synthetic DICOM demo widget
- Schedule Demo / Contact form
- Market opportunity: \$45B imaging market
- LB default backend — no IAP required
- www + apex domains on SSL cert v3

Automated Processing Pipeline routing rules define which steps are required → pipeline dispatches each step in dependency order, hands-free

1. Classify

fills modality
body_part

2. PHI Scan

OCR pixels
flag burned-in text

3. Protocol

TR / TE / flip
vs template

4. Deface

remove face
head imaging only

5. QC Check

SNR · coverage
slice consistency

6. BIDS

NIFTI + sidecar
BIDS structure

7. Export Fwd

STOW-RS or
DIMSE forward

Implementation Phases

Phase 1: Foundation ✓

Terraform (GCP + AWS) · Go API · Upload Portal · Admin Dashboard · PostgreSQL · CI/CD

Phase 2: Processing ✓

Defacing · PHI Detection · QC · BIDS · Classification · Protocol · DIMSE Receiver

Phase 3: Operations ✓

Routing rules · Institutions · Audit · Export Portal · Shares · Email · MCP Server

Phase 4: Production ✓

GCP live (aegis-prod) · Landing Page · Batch import · Synth MRI sidecar (8th) · RBAC

Phase 5: Enterprise 🚧

AEGIS AI Agent · Observability dashboard · Multi-tenant federation · Repo split

Key Open-Source Dependencies

- Go:** suyashkumar/dicom · pgx · testcontainers-go · testify
- Browser:** dcmjs · dicomParser · OHIF Viewer (MIT) · React 19 · Vite
- Defacing:** mri_reface · DeepDefacer · mri_deface · dcm2nix · nibabel
- PHI/OCR:** pytesseract · Google Cloud Vision · AWS Textract · Pillow
- QC/BIDS:** pydicom · numpy · dcm2nix · pynetdicom (C-STORE SCP)
- Synth:** numpy · pydicom · MONAI Generative · torch · huggingface-hub
- Infra:** Terraform Google + AWS providers · Docker distroless/slim

Multi-Cloud + Test Coverage

- GCP:** Cloud Run · Cloud SQL · GCS · IAP · Cloud Armor · KMS · Pub/Sub
- AWS:** ECS Fargate · RDS · S3 · ALB + Cognito · ACM · Secrets Manager
- Local:** Docker Compose · PostgreSQL 15 · Mailpit · local filesystem
- Auth:** GCP IAP · Azure AD Easy Auth · AWS ALB+Cognito · dev auto-auth
- Storage:** STORAGE_MODE=gcs | s3 | local — same Go API, no code changes
- CI:** Go tests (273+) · Python tests (244+) · TS typecheck · Docker (9)
- Domains:** aegisimaging.ai · www · api · admin — SSL cert v3